The Intrinsic Dimensionality of the Stock Market: A Geometric Approach to Factor-Based Market Analysis

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by

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CERTIFICATE

*This is to certify that the work contained in this thesis entitled "**The Intrinsic Dimensionality of the Stock Market: A Geometric Approach to Factor-Based Market Analysis**" is a bonafide work of **Rohan Kumar Mahato** (**Roll No. 220101088**), carried out in the Department of Computer Science and Engineering, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.*

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Chapter 1

Introduction and Motivation

1.1 Background of the Indian Equity Market

The Indian equity market is growing really fast and it is one of the most active emerging markets right now. By 2020, the National Stock Exchange had over 5,000 companies listed on it, and the NIFTY 500 index alone covered about 93% of the total market cap. Since it has companies from so many sectors like IT, pharma, and infra, it basically gives a good picture of how the Indian economy is doing.

Every day, trading in all these stocks generates a huge amount of data. The prices don't just move randomly because they are affected by factors like interest rates, GDP growth, and global trends along with company news. So if we want to understand how the market behaves, we need to study all these stocks together instead of looking at them one by one. This is quite important for people like portfolio managers or risk analysts.

But the main problem is just how big the data is. When you deal with 300 to 400 stocks and continuous data, things get very messy. A lot of traditional models fail because the covariance matrices get noisy and the results don't work well when you test them on new data.

1.2 The Curse of Dimensionality in Financial Markets

In high-dimensional spaces, the usual statistical ideas we learn just don't work well [BHK20]. People call this the "Curse of Dimensionality" and it shows up in a few ways when we look at financial data:

- Pairwise correlations explode: For $N = 400$ stocks, the covariance matrix has $\frac{400 \times 399}{2} \approx 79,800$ unique parameters.
- Distance concentration: In high dimensions, the distance between any two random points becomes almost the same, so clustering methods don't work properly [Vem04].
- Noise dominance: Sample covariance matrices often get dominated by noise instead of showing the real economic signal.

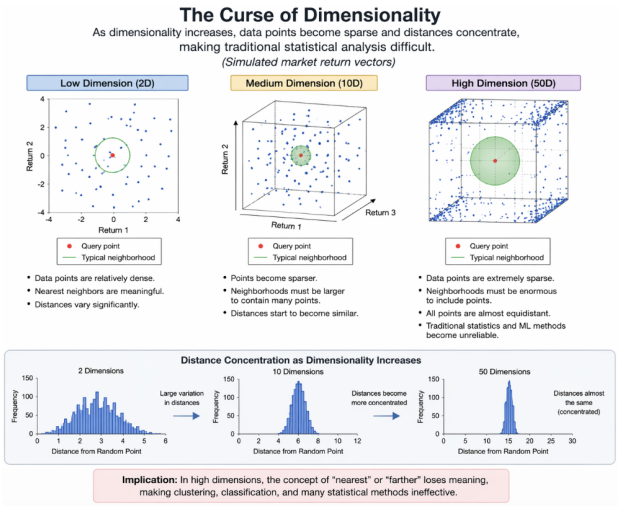


Fig. 1.1 Illustration of the Curse of Dimensionality. As dimensionality increases, data points become increasingly sparse and distances concentrate, making traditional statistical analysis difficult. (Simulated market return vectors)

Random Matrix Theory (RMT) is a great way to separate the actual signal from the noise in these matrices. Some famous papers [LCBP99, PGR+99] showed that a lot of eigenvalues in stock return matrices follow the Marchenko-Pastur distribution which is basically pure noise. This approach has been used for the Indian market too [KKK20].

1.3 Geometric Perspective: Markets as High-Dimensional Objects

In my case, I have looked at the market from a geometric point of view, inspired by things like the Vector Space Model [SWY75] and modern data science [BHK20].

Instead of thinking of each stock as a separate time series, we can treat the whole market on a trading day as a single point or vector in R^N, where N is the number of stocks. Over many days, the market looks like a cloud of points in this high-dimensional space.

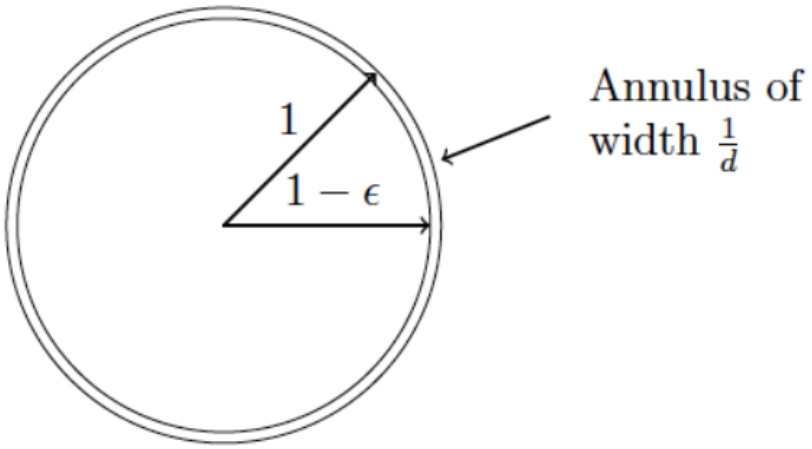


Fig. 1.2 Gaussian Annulus Theorem (Thin Shell Phenomenon). In high dimensions, most of the probability mass of a multivariate Gaussian concentrates in a thin shell at radius $\sqrt{N}$. Purely random markets would lie on this shell; deviations indicate genuine market structure.

This geometric idea is backed by the Gaussian Annulus Theorem. It says that in high dimensions, random points usually lie in a thin shell [BHK20, MU05]. A random market would just look like points on this shell, but real markets show strong patterns because of sectors and economic shifts. These deviations mean there is a simpler structure hidden inside teh high-dimensional space and that is why I wanted to use dimensionality reduction.

1.4 Motivation & Contributions

My main goal for this research is to find that hidden low-dimensional structure in the Indian market and see how it changes over different years and time scales.

This thesis is different from earlier work because of these points:

1. **Rigorous Geometric and Theoretical Foundation**: I didn't just use PCA as a tool but looked at it through Random Matrix Theory and the Johnson-Lindenstrauss Lemma [DG03, Vem04]. This gives us mathematical guarentees that the market structure is actually preserved.

2. **Fixed Stock Universe Approach**: Usually, indices change over time which causes bias. To avoid this, I picked a fixed set of stocks from the NIFTY 500 as of 2004 and tracked them until 2020. This makes the comparison over 15 years much more reliable.

3. **Multi-Frequency Analysis**: We did PCA on daily, weekly, and monthly returns. This helps to see how the market structure changes when we look at short-term noise versus long-term drivers.

4. **Temporal Regime Analysis**: I compared the market across three different periods like 2005–2010 and 2015–2020 to see how economic events changed the statistical factors.

5. **Sectoral Interpretability Focus**: Using average absolute loadings, I tried to quantify which sectors dominate which principal component so the results actually make economic sense.

1.5 Research Objectives

The main things I want to achieve with this thesis are:

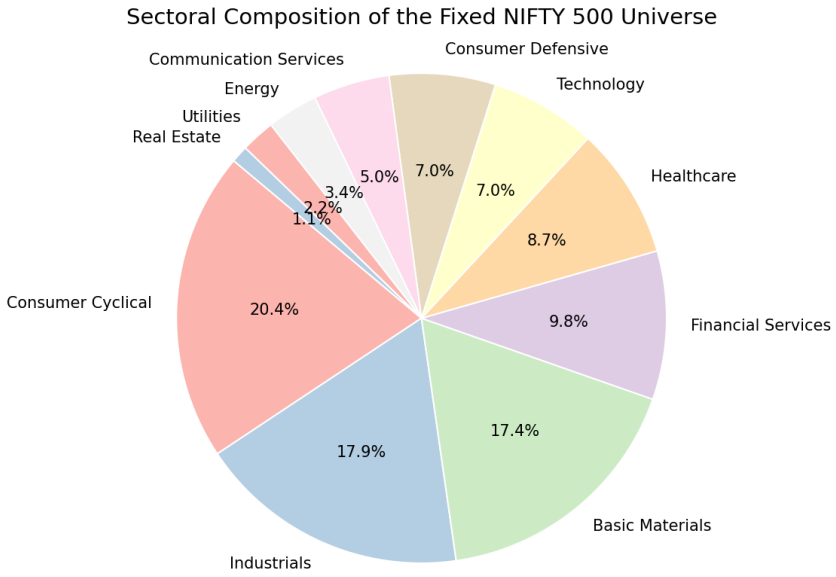


Fig. 1.3 Sectoral Composition of the Fixed NIFTY 500 Universe Used in This Study

- To find the real dimensionality of the Indian market by looking at eigenvalues and variance across different frequencies.
- To build low-rank models of the return matrix that can separate the main risk factors from random noise.
- To check how stable the principal components are during different market phases from 2005 to 2020.
- To measure and explain how different sectors contribute to the main factors we find.
- To see if these results stay the same across daily, weekly, and monthly data.
- To provide some useful ideas for risk management and portfolio building based on classical works [Hul14, Shr04].

1.6 Thesis Organization

The rest of the thesis is organized like this:

Chapter 2 explains the math behind PCA and the geometric theories like Random Matrix Theory.

Chapter 3 covers how we collected the data and cleaned it, including how we handled things like corporate actions for our 2005 stock list.

Chapter 4 talks about the research methodology and how we actually implemented PCA and the subperiod analysis.

Chapter 5 shows the results for the full period from 2005 to 2020 with all the plots and heatmaps.

Chapter 6 is a detailed look at the subperiods to show how the market evolved over time.

Chapter 7 gives the economic meaning of the factors we found and compares them with other studies.

Chapter 8 summarizes everything and talks about the limitations and what can be done in the future.

This chapter basically sets the stage for a deep dive into the hidden structure of the Indian equity market.

Chapter 2

Mathematical Foundations of Dimensionality Reduction

Principal Component Analysis (PCA) is basically the main tool for looking at high-dimensional data and it has deep roots in both old statistics and new data science [BHK20]. In this chapter, I have tried to provide a proper mathematical explanation of PCA by linking the geometric ideas from Chapter 1 with classical linear algebra. We will derive PCA from scratch, look at some theoretical guarentees, and see why it is so useful for financial data.

2.1 From Geometry to Linear Algebra

In the first chapter, we looked at the Indian equity market as a cloud of points in a high-dimensional space R^N. Here, every trading day is a vector and every stock acts as a coordinate axis. Because of the "Curse of Dimensionality" and other phenomena, we know that the real market structure is hidden in a much smaller subspace [BHK20, Vem04].

PCA does exactly what we need because it finds the directions where the data varies the most. This gives us the best linear projection that keeps as much of the geometric info as possible. Basically, for a set of T observations in R^N, PCA looks for an orthonormal basis of $k \ll N$ vectors to minimize teh mean squared reconstruction error.

2.2 Linear Algebra Preliminaries

Let $\mathbf{R} \in R^{T \times N}$ be the matrix of our returns where each row is a trading day and each column is a stock. First, we center the data by subtracting the average of each column to get teh centered matrix $\mathbf{X}$.

The sample covariance matrix $\mathbf{C} \in R^{N \times N}$ is defined as:

$$\mathbf{C} = \frac{1}{T-1}\mathbf{X}^\top \mathbf{X} \tag{2.1}$$

Since $\mathbf{C}$ is symmetric, we can use the Spectral Theorem to get its eigenvalue decomposition:

$$\mathbf{C} = \mathbf{W}\mathbf{\Lambda}\mathbf{W}^\top \tag{2.2}$$

In this case, $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N)$ where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0$. Also, $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_N]$ is an orthogonal matrix where the columns are eigenvectors.

2.3 Derivation of Principal Component Analysis

The first principal component is simply the unit vector $\mathbf{w}_1$ that maximizes the variance of the projection:

$$\mathbf{w}_1 = \arg\max_{\|\mathbf{w}\|=1} \text{Var}(\mathbf{X}\mathbf{w}) = \arg\max_{\|\mathbf{w}\|=1} \mathbf{w}^\top \mathbf{C}\mathbf{w} \tag{2.3}$$

If we use Lagrange multipliers, the optimization problem looks like this:

$$\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^\top \mathbf{C}\mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{w} - 1) \tag{2.4}$$

By differentiating with respect to $\mathbf{w}$, we get the eigenvalue equation:

$$\mathbf{C}\mathbf{w} = \lambda \mathbf{w} \tag{2.5}$$

The solution $\mathbf{w}_1$ for the largest eigenvalue λ_1 gives us the first principal component. We get the next components one by one by adding the condition that they must be orthogonal. This is the same as doing a full spectral decomposition of $\mathbf{C}$ [BHK20].

Another way to do this which is often more stable for computers is by using Singular Value Decomposition (SVD) on the centered data:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \tag{2.6}$$

Here, the right singular vectors in $\mathbf{V}$ are exactly the same as the eigenvectors of $\mathbf{C}$.

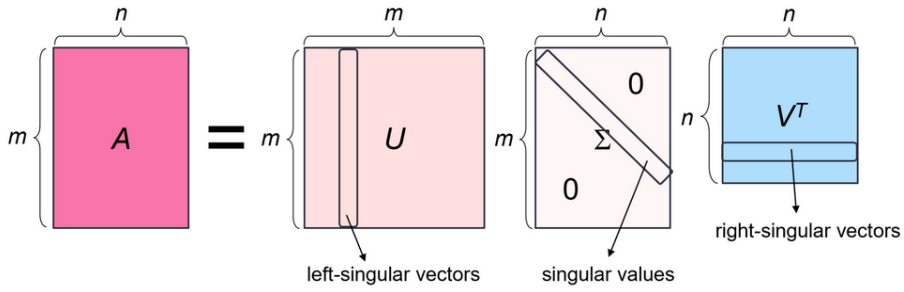


Fig. 2.1 Singular Value Decomposition (SVD) representation of the return matrix. The matrix $\mathbf{V}$ contains the principal component loadings (asset space), $\mathbf{U\Sigma}$ contains the factor scores (time series), and the singular values in $\mathbf{\Sigma}$ quantify the importance of each component.

2.4 Standardization of Returns

Financial returns are a bit tricky and they need careful preprocessing. Different stocks have very different levels of volatility because of their size or sector. If I just used raw returns, the stocks with high volatility would dominate the top components just because of their scale [Hul14].

So, we need to standardize the return series for each stock:

$$\tilde{r}_{t,i} = \frac{r_{t,i} - \hat{\mu}_i}{\hat{\sigma}_i}, \quad i = 1, \ldots, N \tag{2.7}$$

In this formula, $\hat{\mu}_i$ and $\hat{\sigma}_i$ are just the mean and standard deviation for stock i.

After we do this, the covariance matrix becomes a correlation matrix and every stock gets to contribute equally. This is a very important step if we want to get factors that actually make economic sense.

2.5 Low-Rank Approximation and Factor Model

By using only the first k components, we can approximate the standardized return matrix like this:

$$\tilde{R} \approx \mathbf{F}_k \mathbf{W}_k^\top \tag{2.8}$$

Here, $\mathbf{W}_k$ has the first k eigenvectors and the factor score matrix is:

$$\mathbf{F}_k = \tilde{\mathbf{R}} \mathbf{W}_k \in R^{T \times k} \tag{2.9}$$

We can calculate how much variance we explained using this ratio:

$$\text{Explained Variance Ratio} = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{N} \lambda_j} \tag{2.10}$$

If there is no more structure left, the leftover part or teh residual matrix $\mathbf{E} = \tilde{\mathbf{R}} - \mathbf{F}_k \mathbf{W}_k^\top$ should follow the Marchenko-Pastur law from Random Matrix Theory [LCBP99, PGR+99].

2.6 Financial Interpretation of Principal Components

In the stock market, these principal components usually have a clear economic meaning:

- **PC1 (Market Factor):** This one usually has positive loadings for almost all stocks. It basically represents the overall market movement and general economic conditions.
- **PC2 – PCk (Sector/Rotation Factors):** These are orthogonal to PC1 and they

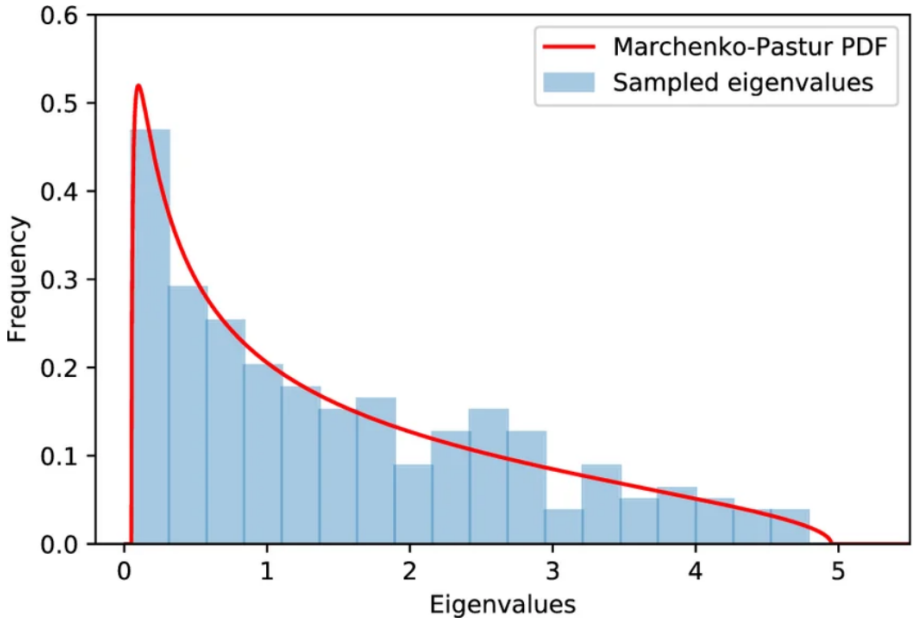


Fig. 2.2 Typical eigenvalue spectrum of a financial correlation matrix. The bulk follows the Marchenko-Pastur distribution (noise), while a few large eigenvalues (signal) deviate significantly.

usually show the difference between sectors like IT versus Banking or Defensive versus Cyclical stocks.

- **Remaining Components:** These are mostly just random noise or effects that only touch a few stocks.

This way of breaking things down fits very well with the Arbitrage Pricing Theory (APT) which is used a lot in finance [Shr04, Hul14].

2.7 Comparison with Other Dimensionality Reduction Methods

Method	Variance Maximization	Orthogonality	Interpretability
PCA	Yes (Total Variance)	Yes	High
Factor Analysis	No (Common Variance)	Yes (with rotation)	Medium
ICA	No (Independence)	No	High (if successful)
t-SNE / UMAP	No (Local Structure)	No	Low (non-linear)
Autoencoders	No (Reconstruction)	No	Low (black-box)

Table 2.1 Comparison of popular dimensionality reduction techniques

Even though there are non-linear methods like autoencoders that can find complex patterns, I think PCA is better for finance. It is faster to compute, easy to reproduce, and it has a very strong theoretical background from linear algebra and Random Matrix Theory [MU05].

2.8 Connection to Random Matrix Theory

By combining PCA with Random Matrix Theory [LCBP99, PGR+99], we can actually tell which eigenvalues are real signals and which ones are just noise. This foundation along with the Johnson-Lindenstrauss Lemma [DG03, Vem04] makes us sure that the components we extracted for this thesis are reliable.

So, this chapter covers all the math we need. In the next few chapters, I will use this theory to look at the actual data from the Indian market.

Chapter 3

Data Collection, Universe Construction and Preprocessing

This chapter provides a comprehensive and transparent description of the data acquisition, cleaning, and preprocessing pipeline developed for this study. A major contribution of this thesis is the construction of a **fixed stock universe** covering the period **2005–2020**, which eliminates survivorship bias and enables consistent cross-period comparison.

3.1 Construction of the Fixed NIFTY 500 Universe

To ensure temporal consistency in the dimensionality of the dataset, a fixed set of stocks was defined based on their membership in the NIFTY 500 index as of the end of 2004.

The official NSE NIFTY 500 inclusion-exclusion history was obtained from the file `IndexInclExcl.xls`. The following procedure was implemented:

1. Load and sort all inclusion/exclusion events by `Event Date`.
2. Filter events up to and including **31 December 2004**.
3. Initialize an empty set. Add company names on "Inclusion" events and remove them on "Exclusion" events.

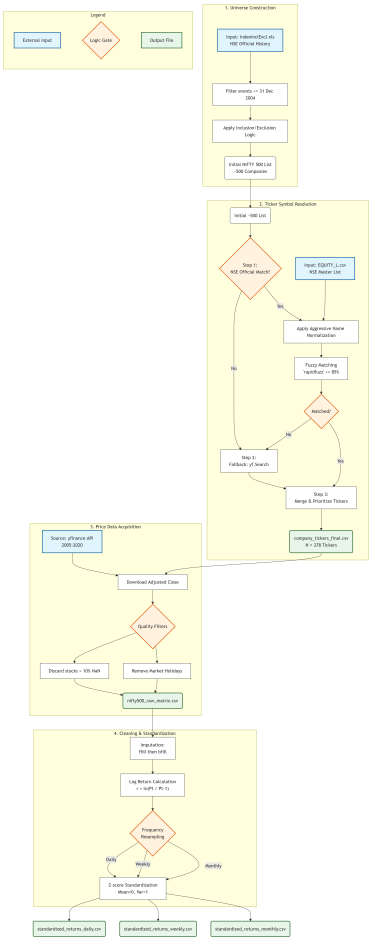


Fig. 3.1 Complete Data Pipeline Flowchart

4. Export the final list of companies present in the index at the end of 2004.

This process yielded **approximately 500 companies** that formed the initial universe.

3.2 Ticker Symbol Resolution and Mapping

Mapping company names to tradable Yahoo Finance tickers is non-trivial due to name variations, mergers, and delistings. A robust multi-source approach was adopted:

3.2.1 Step 1: NSE Official Symbol Matching

- The latest NSE equity master list can be downloaded from:

  `https://archives.nseindia.com/content/equities/EQUITY_L.csv`

- Normalization: Applied aggressive name normalization (removal of suffixes like Ltd, Limited, Corp, etc.).
- Fuzzy Matching: Used the `rapidfuzz` library with a token sort ratio ($\geq 85\%$) for matching.

3.2.2 Step 2: Yahoo Finance Search

For companies not matched via NSE, `yf.Search()` was used with cleaned company names, preferring symbols ending in `.NS`.

3.2.3 Step 3: Merging and Prioritization

NSE-derived tickers were given priority. The final ticker list was saved as **company_tickers_final.csv**

Stage	Companies	Notes
Initial NIFTY 500 (end-2004)	≈ 500	From official inclusions/exclusions
After NSE fuzzy matching	≈ 420	High-confidence matches
After yfinance search	≈ 380	Additional recoveries
Final merged universe	**378**	Used in this study

Table 3.1 Summary of Ticker Mapping Process

3.3 Price Data Acquisition

Historical adjusted closing prices were downloaded using the `yfinance` library for the period **01 January 2005** to **31 December 2020**.

The download function implemented the following quality filters:

- Stocks with more than **10%** missing data were discarded.
- Rows where all stocks had missing values (market holidays) were removed.

This resulted in a final raw price matrix of dimensions $T \times N$, where:

- $N = 378$ stocks

- $T \approx 3,720$ trading days (daily)

The raw price matrix was saved as `nifty500_raw_matrix.csv`.

3.4 Data Cleaning and Imputation

Missing values in the price series were handled using a professional two-step imputation:

1. Forward fill (`ffill`) — carries forward the last observed price.
2. Backward fill (`bfill`) — fills any remaining initial gaps.

This approach preserves the integrity of corporate actions already incorporated in adjusted prices.

3.5 Return Calculation Across Multiple Frequencies

Three frequency datasets were generated to enable robustness analysis:

3.5.1 Daily Returns

$$r_{t,i} = \frac{1}{\sqrt{\Delta t}} \ln\left(\frac{P_{t,i}^{\text{adj}}}{P_{t-1,i}^{\text{adj}}}\right)$$

where Δt is the number of calendar days between trading days (usually 1, but accounts for weekends/holidays).

3.5.2 Weekly Returns

Resampled to Friday-end using `.resample('W-FRI').last()`, followed by simple log returns.

3.5.3 Monthly Returns

Resampled to month-end using `.resample('ME').last()`, followed by log returns.

3.6 Standardization

Each stock's return series was standardized individually to zero mean and unit variance:

$$\tilde{r}_{t,i} = \frac{r_{t,i} - \mu_i}{\sigma_i}$$

This critical step ensures that PCA captures co-movement structure rather than volatility differences.

The final standardized return matrices were saved as:

- `standardized_returns_daily.csv`
- `standardized_returns_weekly.csv`
- `standardized_returns_monthly.csv`

3.7 Final Dataset Characteristics

Frequency	Observations (T)	Stocks (N)	Total Elements
Daily	3,720	378	1,406,160
Weekly	835	378	315,630
Monthly	192	378	72,576

Table 3.2 Final Dataset Dimensions

3.8 Summary

This chapter described a rigorous, reproducible, and bias-minimized data construction process. The fixed-universe, multi-frequency, and carefully cleaned dataset forms a solid foundation for the PCA analysis presented in subsequent chapters. All code used for data collection and preprocessing is available upon request for full reproducibility.

This meticulous preprocessing distinguishes the present work from many existing studies that rely on dynamic index constituents or poorly cleaned data.

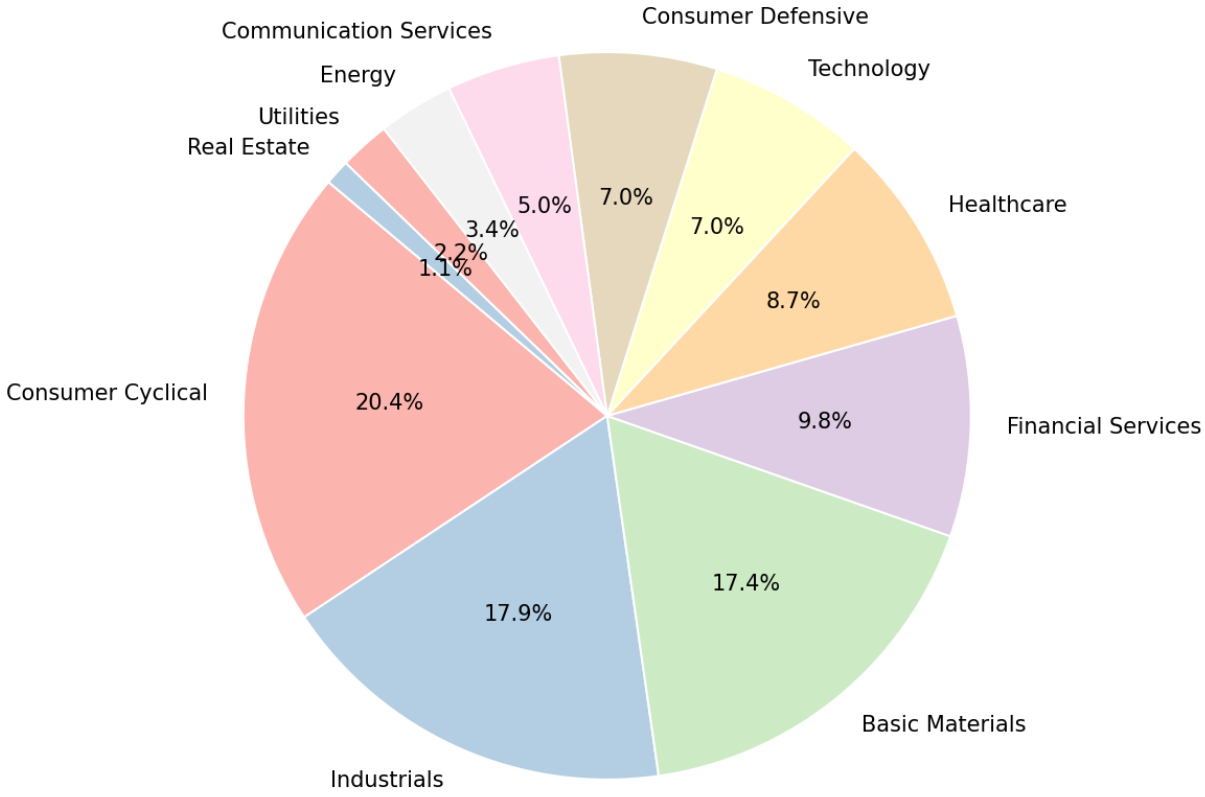


Fig. 3.2 Sector-wise Distribution of the Final 378 Stocks

Chapter 4

Research Methodology

This chapter outlines the complete methodological framework employed in the study. Building upon the mathematical foundations presented in Chapter 2 and the carefully constructed dataset described in Chapter 3, we detail the steps for standardization, principal component extraction, multi-frequency analysis, subperiod investigation, and sectoral interpretation.

4.1 Overview of the Analytical Framework

The core objective of this research is to uncover the latent low-dimensional structure of the Indian equity market using Principal Component Analysis (PCA). The methodology is designed to be robust, reproducible, and economically interpretable. All analyses were performed on the fixed universe of 378 stocks over the period 2005–2020.

The analysis is conducted across three return frequencies (daily, weekly, monthly) and four time windows (full sample + three subperiods), enabling comprehensive robustness checks.

4.2 Return Standardization

As established in Chapter 2, standardization is essential in financial applications of PCA. For each stock i and each frequency, the raw returns $r_{t,i}$ are transformed into standardized returns $\tilde{r}_{t,i}$ as follows:

$$\tilde{r}_{t,i} = \frac{r_{t,i} - \hat{\mu}_i}{\hat{\sigma}_i}, \quad \forall i = 1, \ldots, N \tag{4.1}$$

where $\hat{\mu}_i$ and $\hat{\sigma}_i$ are the sample mean and sample standard deviation of returns for stock i over the analysis period.

This z-score transformation ensures:

- Each stock contributes equally to the total variance.
- The resulting covariance matrix is equivalent to the correlation matrix.
- Results reflect genuine co-movements rather than volatility scaling effects.

Standardization is performed independently for each frequency and each subperiod.

4.3 Principal Component Analysis Procedure

Let $\tilde{\mathbf{R}} \in R^{T \times N}$ denote the matrix of standardized returns. The sample covariance (correlation) matrix is computed as:

$$\mathbf{C} = \frac{1}{T-1} \tilde{\mathbf{R}}^\top \tilde{\mathbf{R}} \in R^{N \times N} \tag{4.2}$$

The eigenvalue decomposition of $\mathbf{C}$ is performed using the symmetric eigen-solver:

$$\mathbf{CW} = \mathbf{W\Lambda} \tag{4.3}$$

where:

- $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N)$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0$,
- $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_N]$ is the matrix of orthonormal eigenvectors.

The principal components are ordered by descending eigenvalues.

4.4 Factor Scores and Low-Rank Approximation

The factor score matrix (principal component time series) for the first k components is obtained by:

$$\mathbf{F}_k = \tilde{\mathbf{R}}\mathbf{W}_k \in R^{T \times k} \tag{4.4}$$

The low-rank approximation of the standardized return matrix is then given by:

$$\tilde{\mathbf{R}} \approx \mathbf{F}_k \mathbf{W}_k^\top \tag{4.5}$$

The proportion of total variance explained by the first k components is:

$$\text{Cumulative Variance Explained} = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{N} \lambda_j} \times 100\% \tag{4.6}$$

4.5 Multi-Frequency Analysis

To examine the robustness and horizon dependence of market structure, PCA is performed independently on three return frequencies:

- **Daily**: Captures short-term dynamics and market microstructure.
- **Weekly**: Balances noise reduction with reasonable sample size.
- **Monthly**: Emphasizes macroeconomic and long-term trends.

All subsequent analyses (scree plots, variance explained, heatmaps) are compared across these frequencies.

4.6 Subperiod Analysis

To study the evolution of market structure over time, the full sample (2005–2020) is divided into three non-overlapping subperiods:

- Period 1: 2005–2010 (Global Financial Crisis and recovery)
- Period 2: 2010–2015 (Post-crisis growth and policy reforms)
- Period 3: 2015–2020 (Demonetization, GST, COVID-19 onset)

PCA is executed separately on each subperiod using the same fixed stock universe. This design allows direct comparison of eigenvalue spectra, explained variance, and sectoral loadings across different economic regimes.

4.7 Sector-Wise Loading Analysis

To facilitate economic interpretation, we quantify sectoral contributions to each principal component. For sector m and component j, the average absolute loading is defined as:

$$L_{m,j} = \frac{1}{|\mathcal{S}_m|} \sum_{i \in \mathcal{S}_m} |w_{i,j}| \tag{4.7}$$

where $\mathcal{S}_m$ is the set of stocks belonging to sector m, and $w_{i,j}$ is the loading of stock i on component j.

This metric identifies which sectors most strongly influence each principal component. Sectoral dominance heatmaps are generated for the first 10 principal components across all frequencies and periods.

4.8 Visualization and Evaluation Tools

The following visualizations and metrics are used to evaluate results:

- **Scree Plots**: Eigenvalue decay (linear and log scale) to identify the "elbow" point.
- **Cumulative Variance Plots**: To determine the number of components needed to explain 70%, 75%, 80%, and 85% of total variance.
- **Stock Loading Heatmaps**: Top stocks versus first 30 principal components.
- **Sectoral Heatmaps**: Average absolute loadings by sector.

Metric	Purpose	Interpretation
Eigenvalue λ_j	Importance of component j	Higher = more variance captured
Cumulative Variance	Dimensionality assessment	Target: 70–85%
Average Abs. Loading $L_{m,j}$	Sectoral dominance	Identifies economic drivers

Table 4.1 Key Evaluation Metrics

4.9 Implementation Details

All computations were performed in Python 3 using:

- `pandas` and `numpy` for data handling,
- `numpy.linalg.eigh` for eigenvalue decomposition,
- `seaborn` and `matplotlib` for high-quality visualizations.

The complete pipeline is modular, allowing seamless execution across frequencies and time windows.

This methodology ensures statistical rigor while maintaining strong economic interpretability, forming the backbone of the empirical analysis presented in the following chapters.

Chapter 5

Full-Sample Empirical Evaluation

This chapter presents the complete PCA results for the full 15-year sample period from 2005 to 2020 using the fixed universe of 378 stocks. Analyses were conducted independently on daily, weekly, and monthly standardized returns, allowing direct comparison of market structure across different time horizons. All results are derived from the eigenvalue decomposition of the sample correlation matrix as described in Chapter 4.

5.1 Overview of Full-Sample Dataset

The full-sample standardized return matrices have the following dimensions:

- Daily: $T = 3{,}720$ observations $\times$ $N = 378$ stocks
- Weekly: $T = 835$ observations $\times$ $N = 378$ stocks
- Monthly: $T = 192$ observations $\times$ $N = 378$ stocks

5.2 Eigenvalue Spectra and Scree Plots

The scree plots in Figure 5.1 (linear scale) and Figure 5.2 (log scale) reveal a classic "elbow" pattern characteristic of financial correlation matrices. The first eigenvalue is markedly

larger than the rest, followed by a rapid decay. Daily data exhibits a slightly slower decay due to higher microstructure noise, while weekly and monthly frequencies show sharper concentration in the leading components.

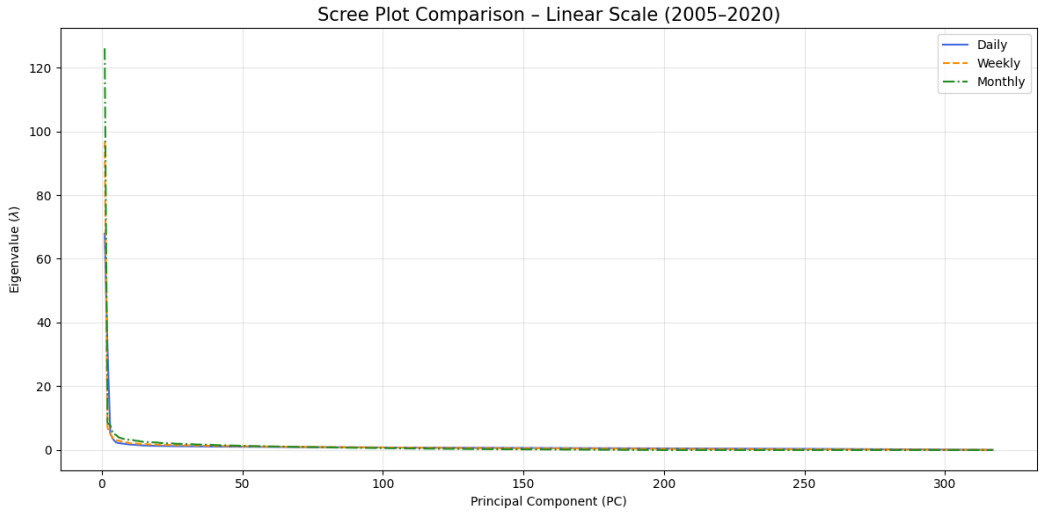


Fig. 5.1 Scree Plot Comparison (Linear Scale) – Full Sample 2005–2020

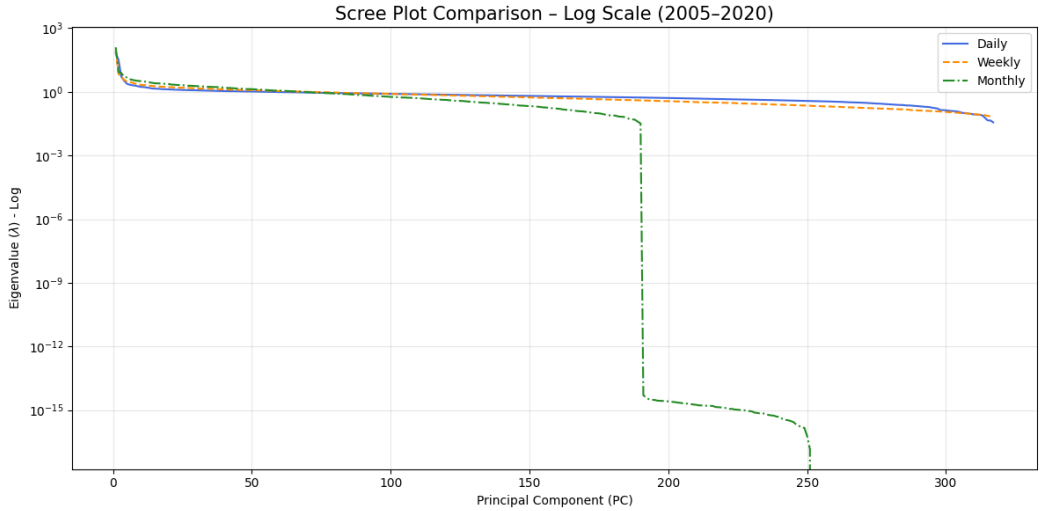


Fig. 5.2 Scree Plot Comparison (Log Scale) – Full Sample 2005–2020

The zoomed scree plot for the first 15 components (Figure 5.3) confirms that the most significant drop occurs within the first 3–5 components, after which eigenvalues approach the noise floor predicted by Random Matrix Theory.

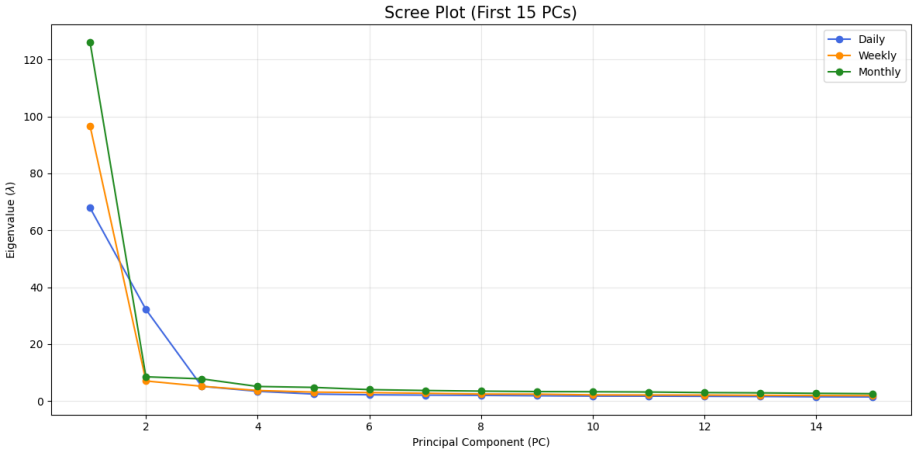


Fig. 5.3 Scree Plot (First 15 Principal Components) – Full Sample 2005–2020

5.3 Cumulative Variance Explained

The cumulative variance plots (Figures 5.4 and 5.5) quantify the dimensionality reduction achieved.

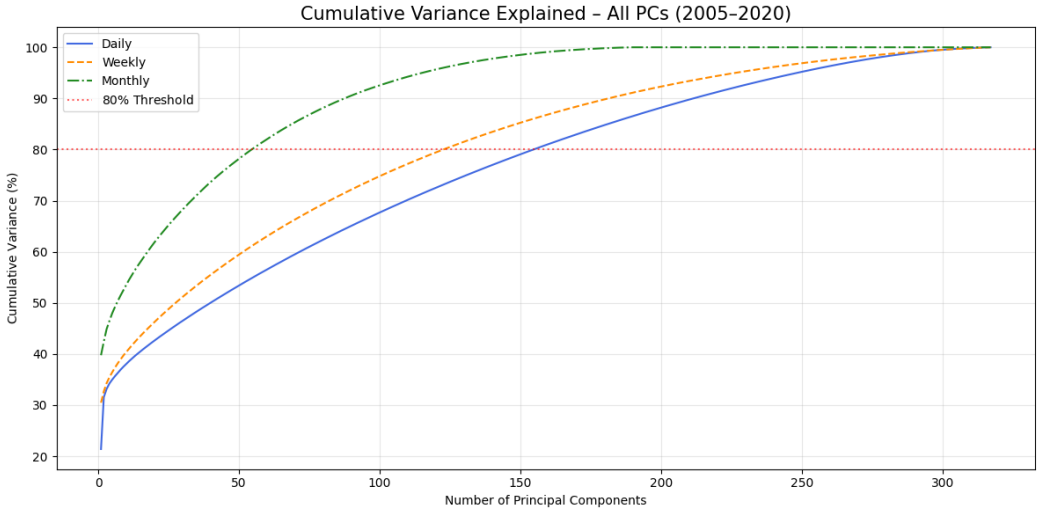


Fig. 5.4 Cumulative Variance Explained (All Principal Components) – Full Sample 2005–2020

Key observations from the variance threshold analysis are summarized in Table 5.1:

Monthly returns require the fewest components to reach any given variance threshold, indicating cleaner, more structured long-term market dynamics. Daily data, conversely, requires nearly twice as many components, reflecting the presence of substantial high-

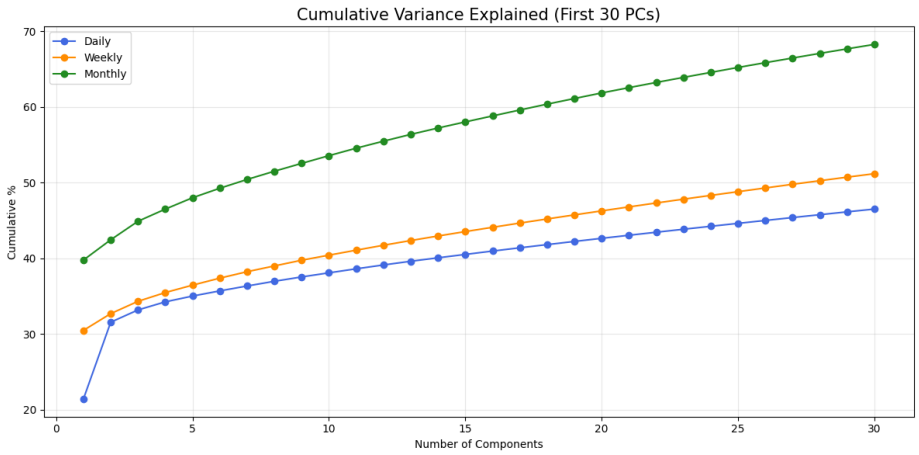


Fig. 5.5 Cumulative Variance Explained (First 30 Principal Components) – Full Sample 2005–2020

Target Variance	Daily	Weekly	Monthly
70%	110 components	83 components	34 components
75%	131 components	101 components	43 components
80%	155 components	123 components	55 components
85%	182 components	149 components	69 components

Table 5.1 Number of Components Required to Explain Target Variance – Full Sample 2005–2020

frequency noise.

5.4 Interpretation of Leading Principal Components

Across all frequencies, the first principal component (PC1) explains approximately 20–40% of total variance and exhibits highly delocalized loadings (positive on nearly all stocks). This is the classic "market factor" driven by broad macroeconomic conditions, rupee movements, global risk appetite, and domestic policy announcements.

PC2 and PC3 (typically explaining 2–10% and 1–5% respectively) capture sectoral rotations:

- Strong positive loadings on Information Technology and Pharmaceuticals (defensive/growth sectors).
- Negative loadings on Financial Services, Energy, and Industrials (cyclical sectors).

Higher-order components (PC4 onward) become increasingly localized and noisy, consistent with the Marchenko-Pastur noise bulk observed in the scree plots.

5.5 Stock-Level Loading Heatmaps

The heatmaps of the top 45 stocks versus the first 30 principal components (Figures 5.6, 5.7, 5.8) reveal a clear hierarchical structure in the return drivers.

The first principal component (PC1) consistently exhibits uniform, positive loadings across the majority of tickers, identifying it as the common "Market Factor" representing systemic risk. In contrast, higher-order components (PC2 through PC30) display localized high-magnitude loadings—both positive and negative—which correspond to sectoral or idiosyncratic shocks.

Notably, as the return frequency decreases from daily to monthly, the loadings become more distinct and less noisy. This transition suggests that while short-term movements are dominated by market-wide volatility, longer-term horizons allow for the emergence of specific structural dependencies and industry-specific clusters within the principal components.

5.6 Sectoral Dominance Heatmaps

Sectoral average absolute loadings for the first 10 components (Figures 5.9, 5.10, 5.11) confirm the economic interpretability of the extracted factors. While **PC1** reflects a broad market factor with relatively uniform loadings across sectors, **Real Estate and Utilities** show significant dominance in **PC2** at the daily frequency. **Financial Services and Information Technology** consistently emerge as primary drivers of **PC3** across frequencies, while **Energy and Consumer Defensive** appear prominently in **PC4**, particularly at the weekly and monthly horizons.

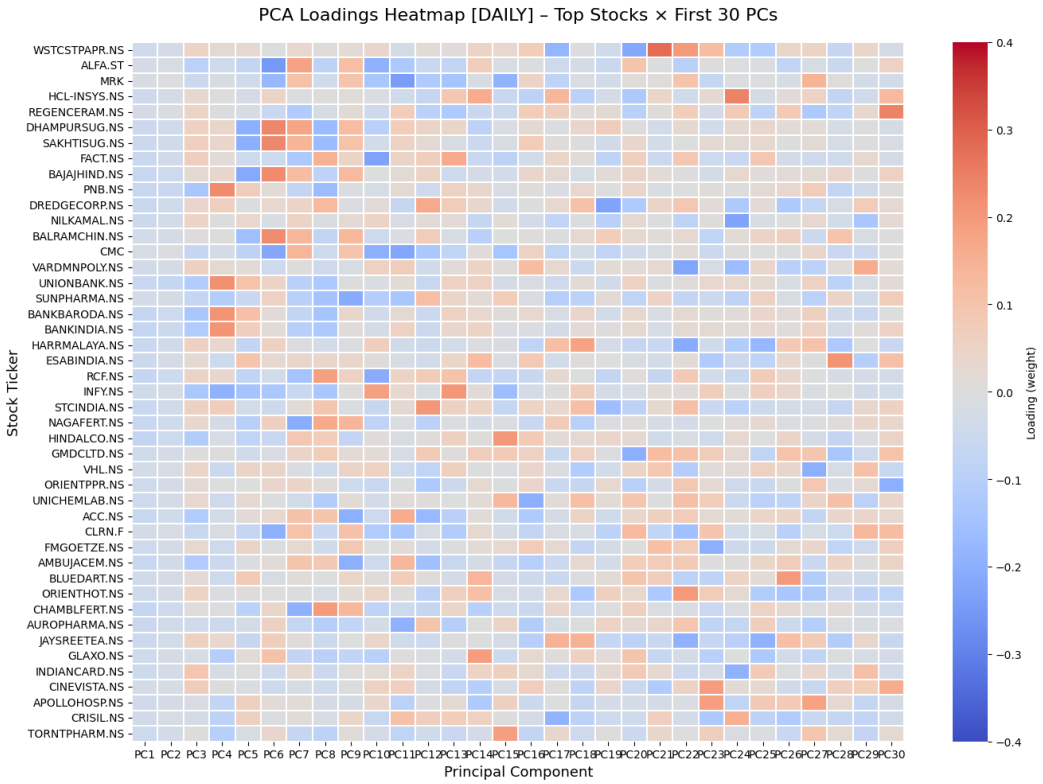


Fig. 5.6 Stock Loading Heatmap (Top Stocks × First 30 PCs) – Daily Returns, 2005–2020

5.7 Comparison Across Frequencies and Economic Insights

The multi-frequency analysis highlights critical horizon-dependent behavior within the Indian equity market:

- **Daily Returns:** These exhibit the highest proportion of idiosyncratic noise and market microstructure effects. The slower eigenvalue decay suggests that daily price movements are influenced by transitory factors such as liquidity constraints and high-frequency news flow, requiring a higher number of components to capture significant total variance.

- **Weekly Returns:** This frequency strikes an optimal balance for PCA, providing a cleaner signal-to-noise ratio. By smoothing out daily volatility, weekly returns allow for the emergence of clearer sectoral dependencies while maintaining a robust sample

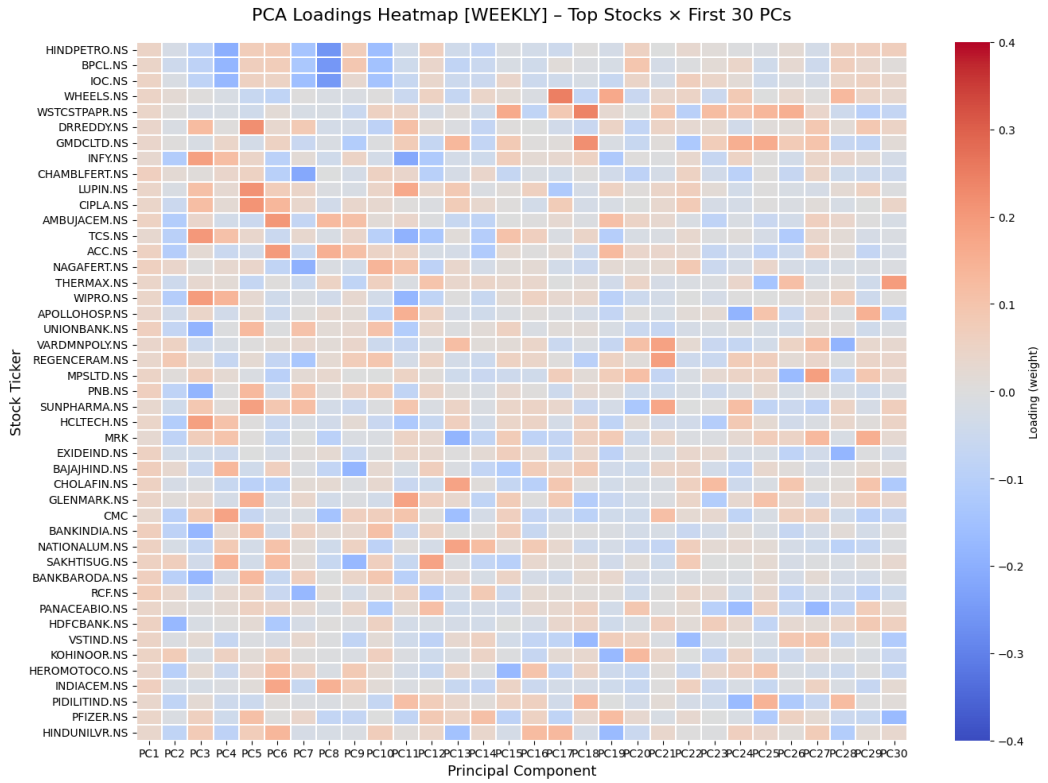


Fig. 5.7 Stock Loading Heatmap (Top Stocks × First 30 PCs) – Weekly Returns, 2005–2020

size for statistical inference.

- **Monthly Returns:** These reveal the underlying macroeconomic structure of the market. Over this horizon, the explanatory power of the leading principal components increases substantially, as returns align with fundamental shifts in economic regimes and systematic risk factors.

These patterns align with financial theory: short-term movements are often dominated by technical and liquidity-driven factors, while longer horizons reflect genuine adjustments to economic fundamentals. The results confirm that the Indian equity market, despite its apparent complexity, possesses a relatively low *intrinsic dimension*. A concentrated set of orthogonal factors (typically 15–30 depending on the frequency) captures the dominant systematic risks, while the remaining dimensions represent diversifiable, idiosyncratic noise.

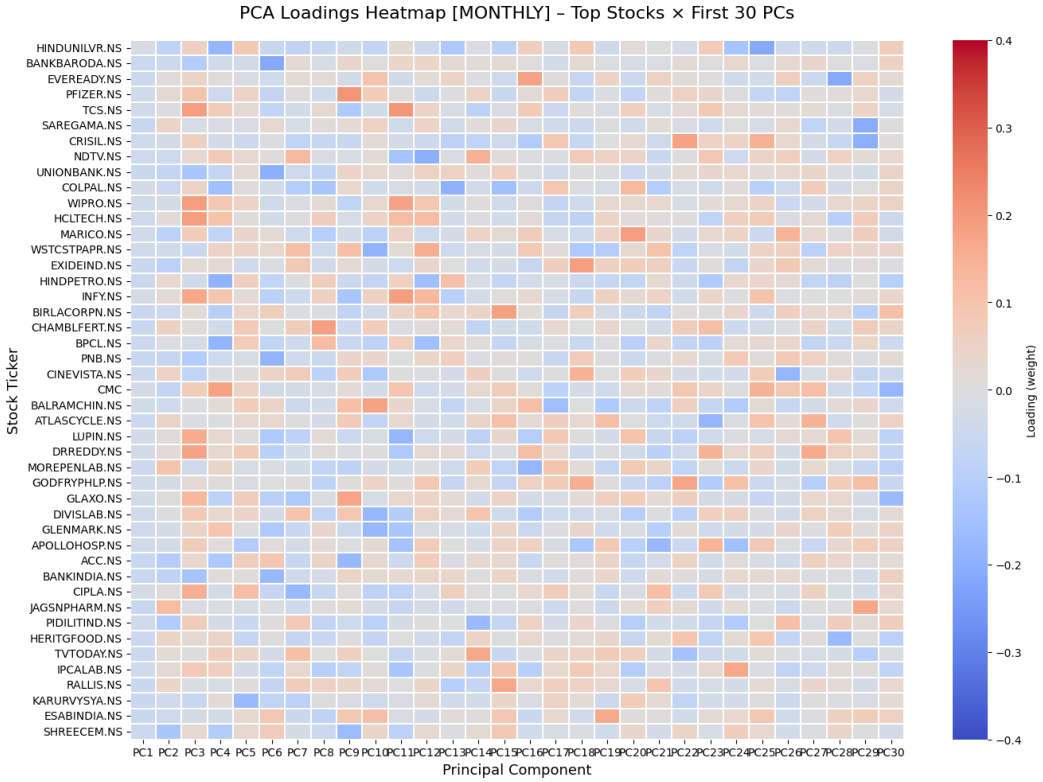


Fig. 5.8 Stock Loading Heatmap (Top Stocks × First 30 PCs) – Monthly Returns, 2005–2020

This full-sample analysis establishes a baseline for understanding market structure. Subsequent chapters will examine how this dimensionality and sectoral dominance evolved across specific economic sub-periods, revealing how the influence of leading components shifted during different phases of market development.

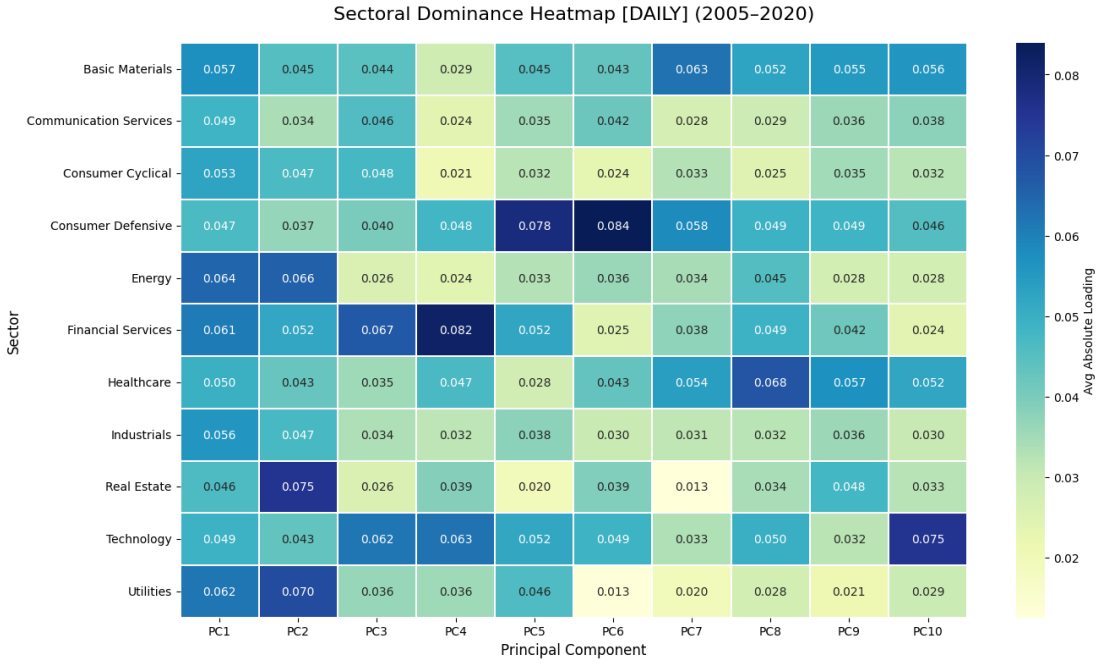


Fig. 5.9 Sectoral Dominance Heatmap – Daily Returns, 2005–2020

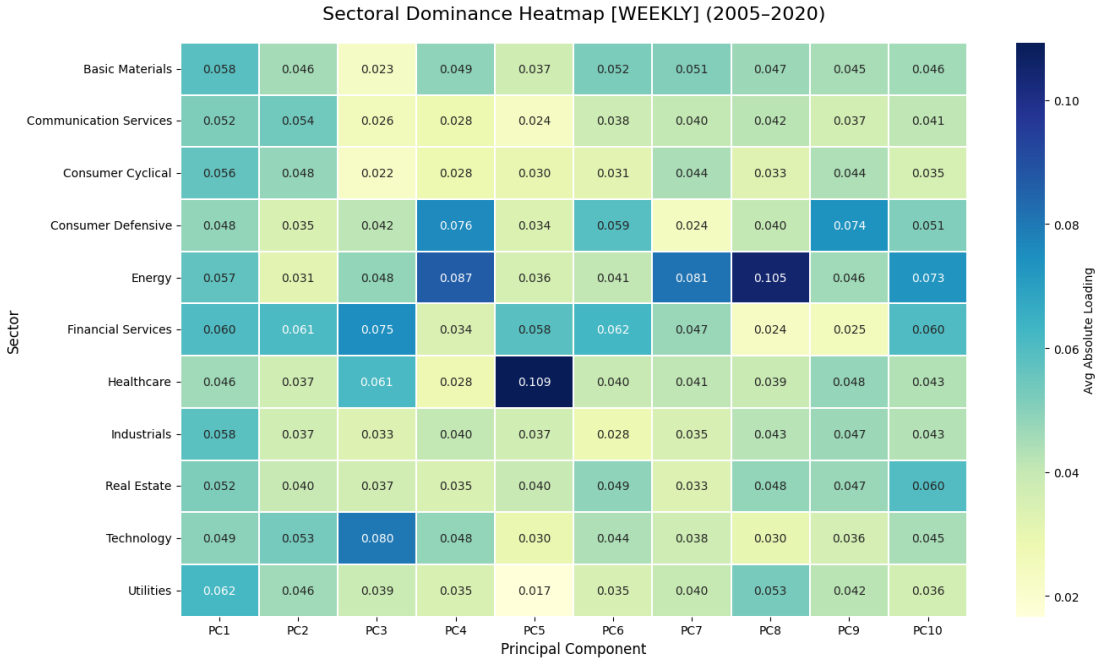


Fig. 5.10 Sectoral Dominance Heatmap – Weekly Returns, 2005–2020

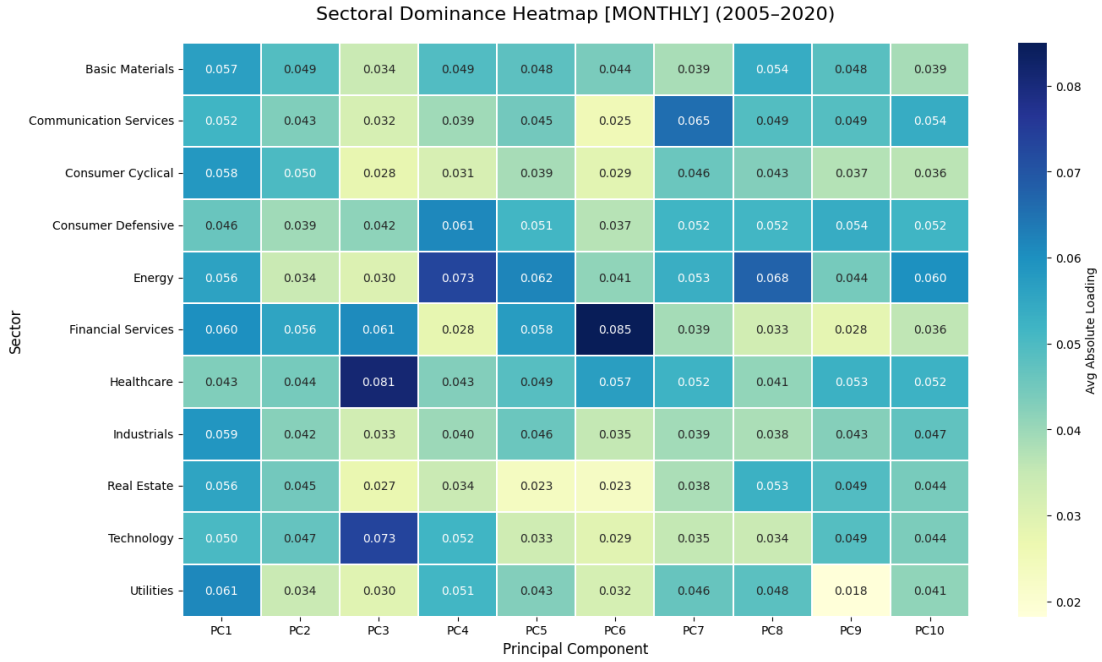


Fig. 5.11 Sectoral Dominance Heatmap – Monthly Returns, 2005–2020

Chapter 6

Subperiod Analysis and Temporal Evolution of Market Structure

This chapter investigates how the latent structure of the Indian equity market evolves over time by partitioning the 2005–2020 period into three distinct economic regimes. Using a fixed universe of 378 stocks, we isolate structural shifts in market factor dominance, sectoral contributions, and intrinsic dimensionality.

6.1 Subperiod Definition and Economic Context

The study period is divided into three five-year windows, each characterized by unique macroeconomic drivers:

- **2005–2010 (Crisis and Recovery)**: Characterized by the pre-2008 "India Shining" bull run followed by the extreme systemic shock of the Global Financial Crisis (GFC). This period tests market resilience and synchronization during a liquidity collapse.
- **2010–2015 (Stabilization and Transition)**: A period of post-crisis stabilization, grappling with high inflation (the "taper tantrum" of 2013) and a significant political transition following the 2014 general elections.

- **2015–2020 (Reform and COVID-19)**: A regime of major structural shifts including Demonetization, the implementation of GST, and the NBFC liquidity crisis, culminating in the unprecedented exogenous shock of the COVID-19 pandemic.

6.2 Comparative Eigenvalue Analysis

6.2.1 2005–2010: Crisis-Driven Market

The PCA results for this period exhibit a "heavy-headed" distribution. The first principal component (PC1) is exceptionally dominant, particularly in monthly returns, where the eigenvalue reaches its highest levels in the study ($\lambda \approx 190$).

The eigenvalue decay is extremely steep, indicating that a vast majority of stock variance was driven by a single systemic factor—the global liquidity cycle. During the 2008 crash, idiosyncratic differences between stocks vanished as the market moved in near-perfect synchronization. As shown in the cumulative variance analysis, the market's "intrinsic dimensionality" was at its lowest, requiring the fewest components to capture the 80% variance threshold.

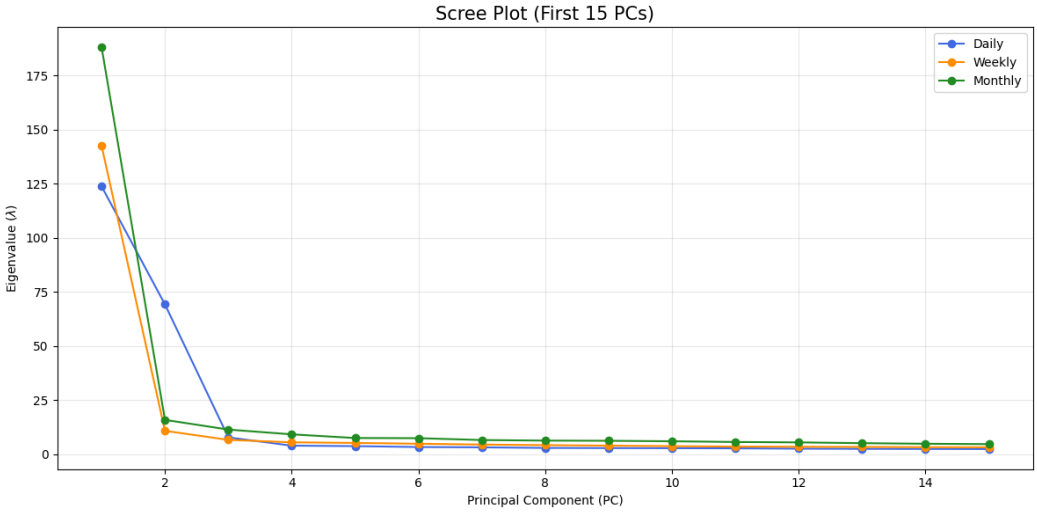


Fig. 6.1 Scree Plot Comparison (Linear Scale) – 2005–2010

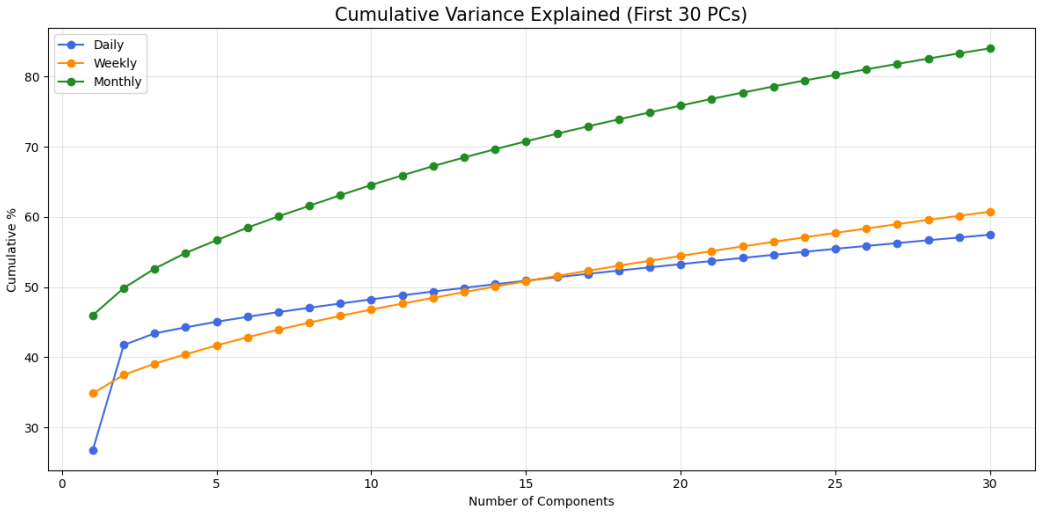


Fig. 6.2 Cumulative Variance Explained (First 30 PCs) – 2005–2010

6.2.2 2010–2015: Transition and Recovery Period

In this subperiod, the dominance of PC1 recedes significantly. The PC1 eigenvalue for monthly data drops to approximately 65, and the scree plot exhibits a much flatter decay. This transition suggests a "decoupling" of stocks from a single systemic driver, allowing for the emergence of PC2 and PC3 as significant secondary factors. These higher-order components capture sectoral divergence as the market focused on domestic policy shifts and recovery rather than global contagion.

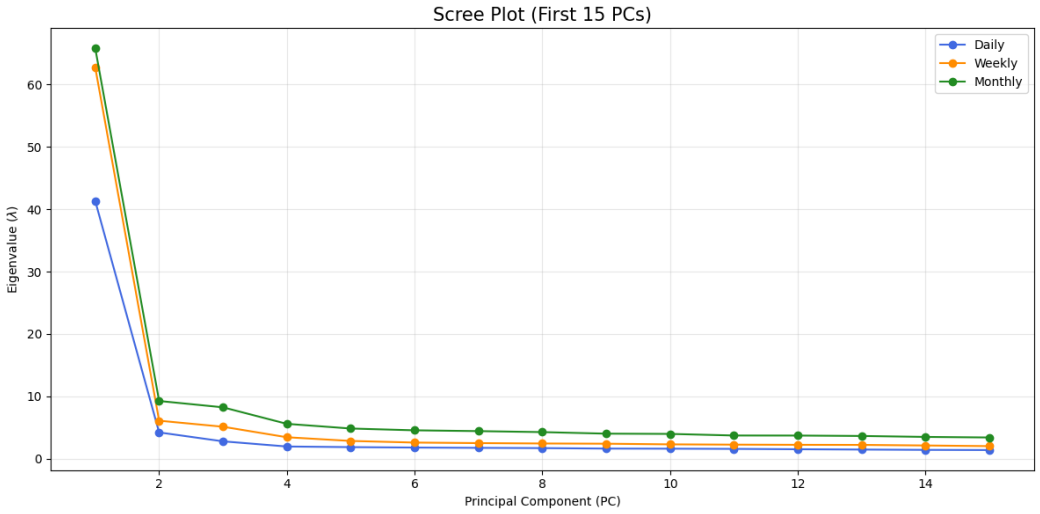


Fig. 6.3 Scree Plot Comparison (Linear Scale) – 2010–2015

6.2.3 2015–2020: Reform and COVID Era

The final subperiod shows a resurgence in PC1 dominance, particularly toward the 2020 pandemic onset. However, the structure is more "thematic" than in 2005. While PC1 captures the COVID-19 shock, PC2 and PC3 exhibit remarkably high and clear loadings in the Healthcare and IT sectors. This suggests that the market structure in this era was defined by a tension between systemic panic (PC1) and specific "winner" sectors (PC2-PC4).

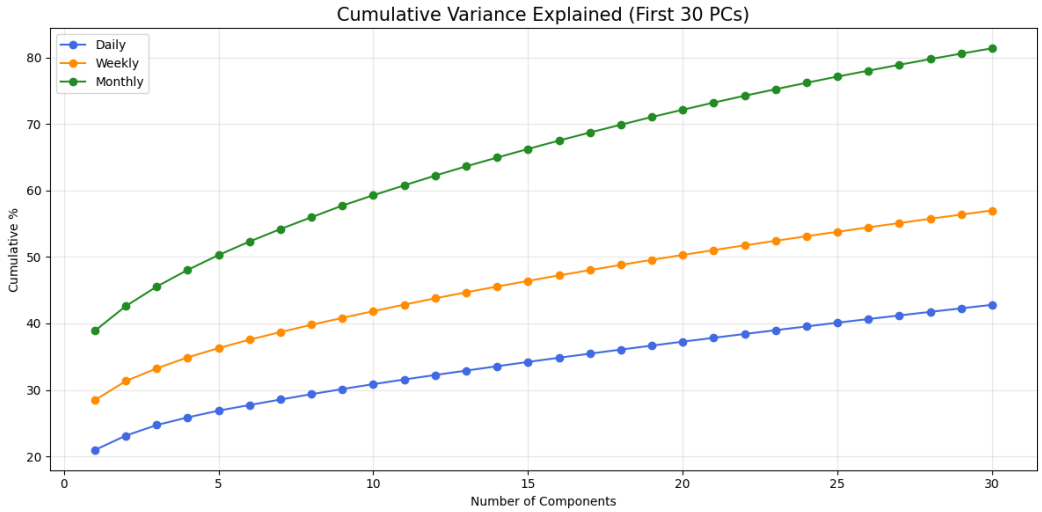


Fig. 6.4 Cumulative Variance Explained (First 30 PCs) – 2015–2020

6.3 Number of Components Required Across Subperiods

The following table quantifies the intrinsic dimensionality of the Indian market across regimes. A lower number of required components indicates a more unified, systematic market, while a higher number indicates a more diverse, idiosyncratic market.

Interpretation: The 2005–2010 period exhibits the highest co-movement (lowest dimensionality), requiring only 25 monthly components for 80% variance. Conversely, the 2010–2015 period is the most diversified, reflecting a quieter macroeconomic environment where individual stock and sector stories could emerge.

Target	2005–2010	2010–2015	2015–2020	Full Sample
70% Variance (Daily)	69	107	101	110
80% Variance (Daily)	112	124	140	155
80% Variance (Monthly)	25	33	29	55

Table 6.1 Number of Components Needed for Target Variance (Daily & Monthly)

6.4 Sectoral Evolution Across Subperiods

The sectoral leadership changed significantly as the economy matured:

- **2005–2010**: Dominance was concentrated in **Financial Services** and **Energy**. During the GFC, the financial sector acted as the primary transmitter of volatility, showing high absolute loadings across the first five components.
- **2010–2015**: A shift toward **Financials** and **Real Estate**. Real Estate emerged as a powerful idiosyncratic cluster (PC8 Monthly), reflecting the domestic credit and property cycles.
- **2015–2020**: The era of the **Healthcare** "super-factor." Healthcare exhibits the highest sectoral dominance recorded in the study (PC3/PC4), particularly as a hedge and response factor during the pandemic.

6.5 Stock Loading Heatmaps Across Subperiods

The stock-level heatmaps confirm that PC1 loadings remain positive and relatively uniform across all periods. However, higher-order components (PC10 and above) show increasingly localized "streaks," indicating that idiosyncratic risks are confined to specific clusters of stocks. This separation between systematic and idiosyncratic risk is most pronounced in monthly data, where the "signal" of sectoral clusters is not obscured by daily noise.

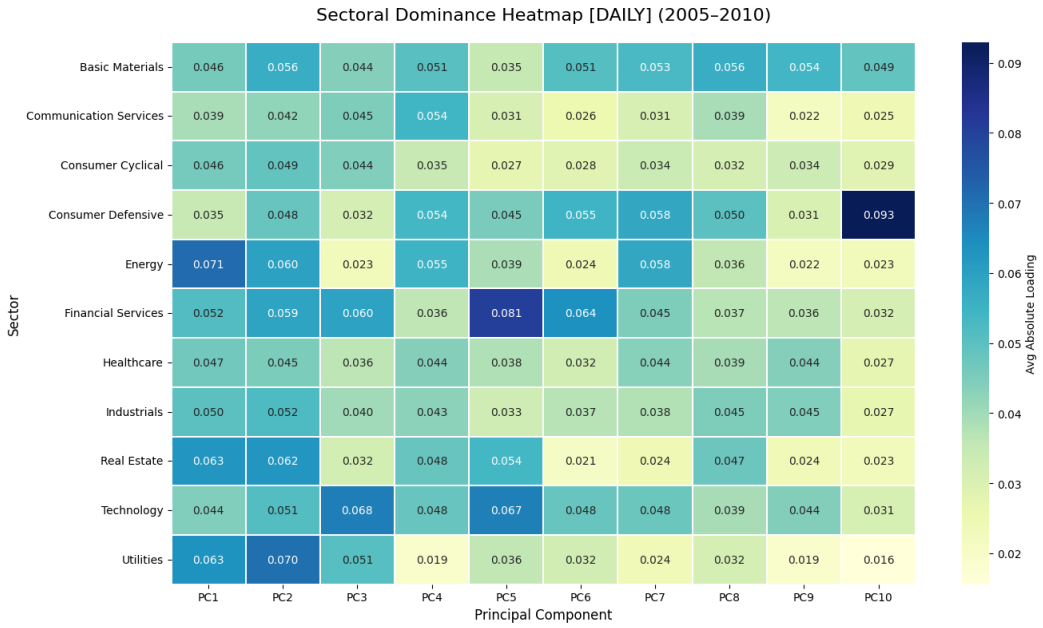


Fig. 6.5 Sectoral Dominance Heatmap (Daily) – 2005–2010

6.6 Discussion: Temporal Evolution of Market Structure

The synthesis of these findings leads to several critical insights:

1. **Crisis Synchronization**: Systemic shocks (2008, 2020) act as "collapsing forces" that reduce market dimensionality, making traditional diversification less effective as PC1 absorbs nearly all variance.

2. **Sectoral Differentiation**: The transition from Energy/Financials (2005) to Healthcare/IT (2020) dominance reflects the maturing of the Indian economy from a commodity/infrastructure focus toward service and growth-oriented sectors.

3. **Signal-to-Noise Ratio**: Monthly returns consistently provide a clearer structural map of the market. The "noise floor" (higher-order eigenvalues) is more stable in monthly data, allowing for better identification of true regime shifts.

4. **Frequency Dependence**: Daily returns are dominated by market microstructure and liquidity flows, which tend to mask structural changes that are only visible at

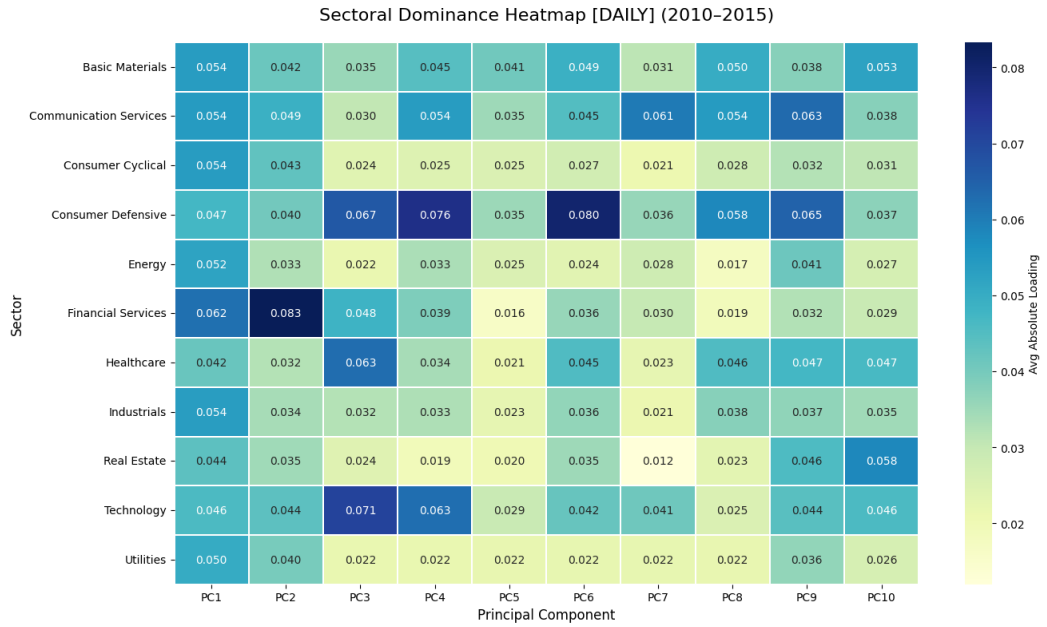


Fig. 6.6 Sectoral Dominance Heatmap (Daily) – 2010–2015

lower frequencies.

6.7 Conclusion

This subperiod analysis demonstrates that the Indian equity market structure is a dynamic entity, deeply intertwined with macroeconomic regimes. While the market factor (PC1) is an ever-present driver, the "flavor" of secondary risks evolves from industrial clusters to service-led growth. For portfolio managers, these results imply that risk models must be frequency-aware and adaptive to the changing "intrinsic dimensionality" of the market.

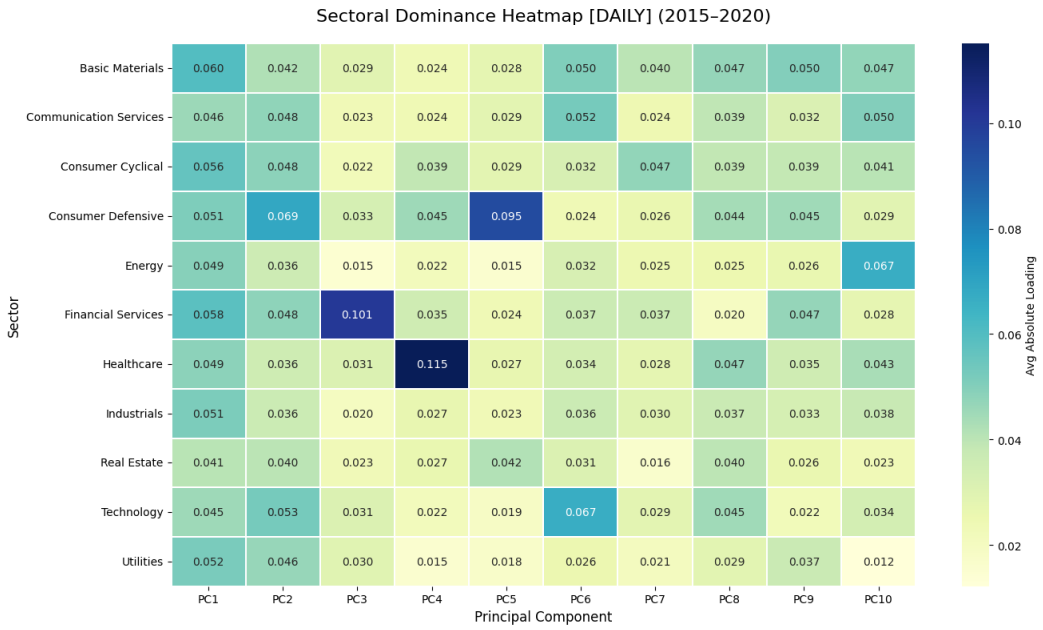


Fig. 6.7 Sectoral Dominance Heatmap (Daily) – 2015–2020

Chapter 7

Economic Interpretation & Market Insights

7.1 Economic Interpretation of Principal Components

The PCA results reveal a hierarchical factor structure where a limited number of orthogonal components capture the primary systematic drivers of the Indian market.

7.1.1 The Market Factor (PC1)

The first principal component (PC1) consistently explains the largest portion of total variance, ranging from approximately 25% (Daily) to over 60% (Monthly) in specific subperiods. Across all frequencies, PC1 loadings are positive and uniform, identifying it as the systemic **Market Factor**.

- **Macroeconomic Drivers**: PC1 serves as a proxy for aggregate shocks including GDP growth expectations, interest rate cycles (RBI policy), and global risk appetite.
- **Temporal Dynamics**: PC1 dominance spikes during crisis periods (2008 and 2020), indicating that systemic risk overrides idiosyncratic sectoral drivers during market stress.

7.1.2 Sectoral and Thematic Factors (PC2–PC5)

Higher-order components represent localized risk clusters.

- **Interpretation**: PC2 and PC3 frequently distinguish between *Cyclical* (Financials, Energy) and *Defensive* sectors (IT, Healthcare). For instance, in the 2015–2020 period, PC3 specifically isolated the Healthcare "super-factor."
- **Frequency Impact**: Interpretation improves at monthly frequencies as high-frequency microstructure noise is filtered, leaving behind cleaner structural dependencies.

7.2 Key Insights on Indian Equity Market Structure

1. **Low Intrinsic Dimensionality**: Despite 378 stocks, the "true" dimensionality is low; 15–30 monthly components capture 80% of variance, suggesting significant redundancy in market movements.
2. **Regime Dependence**: Intrinsic dimensionality collapses during crises (fewer PCs needed) and expands during stable growth regimes (2010–2015), reflecting higher sectoral divergence.
3. **Noise vs. Structure**: Daily returns are dominated by a "noise bulk" (high dimensionality), while monthly returns reveal a highly interpretable macroeconomic backbone.
4. **Market Efficiency**: The presence of a significant "eigenvalue bulk" consistent with Random Matrix Theory (RMT) suggests that a large portion of daily fluctuations is purely idiosyncratic and statistically indistinguishable from noise.

7.3 Robustness Analysis

7.3.1 Multi-Frequency and Subperiod Stability

PCA results show high consistency across weekly and monthly frequencies. While sectoral leadership shifts—moving from Energy/Financials in 2005 to IT/Healthcare in 2020—the *hierarchical structure* (one market factor + sector clusters) remains a robust feature of the Indian market across all 15 years.

7.3.2 Comparison with Random Matrix Theory

Comparing empirical eigenvalues with Marchenko-Pastur bounds confirms that the top 15–40 eigenvalues (depending on frequency) reside outside the "noise bulk." This validates that the extracted factors are statistically significant and represent genuine economic signals rather than random correlations.

7.4 Comparison with Existing Literature

- **Global Alignment**: Our findings align with *Laloux et al. (1999)* and *Plerou et al. (1999)*, confirming that a few leading PCs capture systemic risk while the rest represent a noise bulk.
- **Indian Context**: We extend *Kumar (2020)* by using a fixed-universe approach and multi-frequency analysis, revealing that the Indian market's integration has increased over time, particularly in the monthly macroeconomic dimension.

7.5 Practical Implications

- **Portfolio Construction**: Investors can achieve superior diversification by hedging the first few PCs or by tilting towards sectors identified in the thematic PCs (PC2-PC5).

- **Risk Management**: Covariance matrices can be "cleaned" by retaining only significant eigenvalues and shrinking the noise bulk, leading to more stable optimized portfolios.
- **Regulator Perspective**: Spikes in PC1 variance serve as a real-time indicator of systemic fragility and market-wide panic.

7.6 Limitations of the Study

- **Linearity**: PCA assumes linear relationships; it may underestimate risk during nonlinear "tail events."
- **Fixed Universe**: The requirement for continuous data from 2005–2020 introduces a survivorship bias, potentially overstating market stability.
- **Exogenous Factors**: Factors are latent; PCA identifies *where* risk is concentrated but not the *explicit* cause (e.g., specific policy change).

Chapter 8

Conclusion, Limitations & Next Steps

8.1 Main Takeaways

My analysis of the Indian market leads to a few clear conclusions about its underlying structure:

1. The market has a surprisingly low intrinsic dimensionality, despite having hundreds of stocks, just 15 to 30 components are enough to explain most of the movement.

2. The first principal component (PC1) acts as a universal "Market Factor" that becomes overwhelmingly dominant during crashes, especially in 2008 and 2020.

3. Moving from daily to monthly data significantly cleans up the results, showing that the true economic backbone of the market is often buried under short-term noise.

4. Sectoral clusters like Healthcare and Financials have distinct lives of their own, often appearing as independent risk factors (PC2–PC5) that evolve with the economy.

5. The "noise bulk" identified through Random Matrix Theory suggests that a large portion of daily price changes is essentially random and shouldn't be over-analyzed by investors.

8.2 What This Study Adds

This work contributes to our understanding of emerging markets in a few specific ways:

- Using a fixed universe of 378 stocks over 15 years provides a stable, long-term view that avoids the distortions of changing index compositions.
- The multi-frequency approach proves that market structure looks different depending on whether you are an intra-day trader or a long-term investor.
- It provides a clear mapping of how sectoral dominance shifted from Energy and Infrastructure in the early 2000s to Healthcare and Tech by 2020.

8.3 Applications

The findings here have direct uses for anyone working in the Indian financial space:

- **Portfolio Construction**: Use the top 5 components to build a "diversified" basket that actually covers different economic risks.
- **Risk Management**: Clean up correlation matrices by stripping out the noise bulk to avoid underestimating risk during crises.
- **Index Design**: Evaluate if current benchmarks are truly capturing the sectoral factors that actually drive the market.
- **Regulation**: Monitor the strength of PC1 as an early warning sign for systemic stress and market-wide panic.
- **Investors**: Focus on monthly structural trends rather than daily fluctuations to avoid being misled by statistical noise.

8.4 Acknowledging the Gaps

No model is perfect, and there are a few things this study doesn't cover:

- PCA is a linear tool, so it might miss the complex, non-linear "black swan" events that happen during extreme panics.
- By using a fixed universe, I've focused on survivors, which might make the market seem a bit more stable than it actually was for every company.
- The model identifies patterns but doesn't explicitly tell us *which* macro variable (like inflation or interest rates) caused them.
- I assumed the market structure stays relatively steady within 5-year blocks, but in reality, it changes every single day.

8.5 Looking Ahead

There is still a lot to explore in this area to make these models even more useful:

1. Using "Dynamic PCA" to see how market dimensions shift in real-time rather than using fixed windows.
2. Explicitly linking these PCA factors to external data like RBI policy rates or global oil prices.
3. Testing these same methods across different emerging markets to see if the "Indian structure" is unique.
4. Developing trading strategies that specifically target the gaps between signal and noise identified here.

8.6 Final Remarks

The most important lesson from this research is that the Indian equity market isn't just a random collection of stock prices, it has a deep, evolving structure. While short-term daily moves are mostly noise, the long-term trends are driven by a very small number of powerful economic forces. Understanding that the market's "dimension" changes during a crisis is essential for anyone trying to manage risk in a world that is becoming increasingly connected.

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